

SWAN ISLAND, Australia

WMO: 959850

Lat: 40.729S

Lon: 148.125E

Elev: 51

StdP: 14.67

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	42.2	43.6	29.3	23.4	47.0	32.5	27.0	46.9	41.5	53.8	36.9	53.2	8.5	250	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.2	73.1	66.6	71.1	65.3	69.6	64.1	67.9	71.2	66.7	70.0	65.6	68.7	11.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	98.2	69.5	65.4	93.9	68.6	64.2	90.1	67.8	32.3	71.6	31.4	70.2	30.5	68.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.1	32.3	29.6	38.6	79.5	1.2	3.6	37.8	82.1	37.1	84.2	36.4	86.2	35.6	88.8		
(5)			35.1	69.7	1.4	1.3	34.1	70.6	33.3	71.3	32.5	72.0	31.5	72.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	63.7	64.3	62.9	59.5	55.6	52.6	51.1	51.0	52.6	54.9	57.8	60.8
	DBStd	5.70	3.42	3.18	3.76	3.46	3.37	3.09	2.68	2.72	2.70	3.01	3.32	3.26
	HDD50	63	0	0	0	0	1	10	20	22	8	2	0	0
	HDD65	2935	66	47	86	166	291	372	431	436	373	314	218	135
	CDD50	2684	423	399	400	285	176	88	54	52	85	153	234	335
	CDD65	77	24	26	20	1	0	0	0	0	0	0	1	5
	CDH74	50	24	11	10	0	0	0	0	0	0	0	1	4
	CDH80	4	3	1	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.1	15.5	14.8	14.4	14.1	15.1	14.4	15.3	15.2	15.4	15.3	15.3	15.9
Precipitation	PrecAvg	23.5	1.8	1.2	1.7	2.0	2.4	2.0	2.4	2.5	1.9	1.8	1.9	2.0
	PrecMax	41.2	7.3	5.5	5.6	6.6	8.6	5.5	6.1	9.4	4.9	4.8	3.1	6.3
	PrecMin	14.9	0.2	0.0	0.1	0.1	0.4	0.8	0.4	0.6	0.6	0.3	0.2	0.1
	PrecStd	5.2	1.5	1.2	1.2	1.4	1.7	0.9	1.3	1.7	1.0	1.0	0.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	75.1	75.3	69.7	64.7	61.1	58.3	59.2	62.1	66.0	70.3	73.7
		MCWB	68.4	68.2	68.1	63.6	59.4	56.8	54.1	53.7	55.8	57.8	63.0	65.5
	2%	DB	72.8	72.4	71.8	67.6	62.9	59.4	57.1	57.4	59.5	62.8	66.8	69.5
		MCWB	66.3	66.5	65.8	62.5	58.5	55.6	53.1	53.1	54.6	56.7	61.3	63.3
	5%	DB	70.7	70.7	69.9	65.8	61.7	58.2	56.2	56.1	58.2	61.0	64.4	67.4
		MCWB	65.0	65.2	64.4	60.9	57.6	54.6	52.5	52.2	53.7	55.7	59.6	61.7
	10%	DB	68.9	69.1	68.2	64.6	60.7	57.2	55.4	55.2	57.0	59.6	62.7	65.8
		MCWB	63.5	64.0	63.7	60.1	56.7	53.8	51.8	51.5	52.8	54.7	58.0	60.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.2	69.3	69.1	65.4	61.3	58.7	55.3	54.9	57.2	59.9	64.7	66.6
		MCDB	73.0	72.6	73.2	67.5	62.7	59.9	57.3	56.7	59.3	62.9	67.9	71.2
	2%	WB	67.5	67.5	67.4	63.4	59.8	56.5	54.0	53.9	55.6	58.3	62.1	64.7
		MCDB	71.1	70.9	70.4	66.0	61.8	58.1	56.1	56.1	58.1	61.1	65.9	68.0
	5%	WB	66.2	66.5	66.2	62.3	58.4	55.3	53.2	52.9	54.4	57.0	60.5	63.3
		MCDB	69.7	69.6	69.2	65.0	61.0	57.3	55.4	55.4	57.1	60.0	63.7	66.3
	10%	WB	64.7	65.4	64.8	61.1	57.3	54.2	52.4	52.1	53.4	55.8	59.1	61.8
		MCDB	68.0	68.4	67.7	63.8	60.1	56.6	54.8	54.6	56.1	59.0	62.1	64.8
Mean Daily Temperature Range	5% DB	MDBR	8.7	8.2	8.5	8.0	7.5	6.9	7.0	7.7	8.1	8.5	8.5	8.7
		MCDBR	10.9	9.6	9.7	8.4	7.3	6.7	6.7	7.7	9.0	9.7	10.4	10.2
		MCWBR	6.8	6.4	6.3	6.0	5.8	5.0	5.1	5.6	6.1	6.7	6.7	6.7
	5% WB	MCDBR	10.1	9.0	8.8	7.8	6.5	5.9	5.9	6.8	7.7	9.0	9.8	9.8
		MCWBR	7.2	6.6	5.9	6.4	5.8	4.8	5.0	5.5	6.0	6.6	6.8	7.2
Clear-Sky Solar Irradiance	taub	0.349	0.345	0.335	0.321	0.308	0.302	0.300	0.312	0.333	0.335	0.359	0.344	
	taud	2.530	2.559	2.581	2.603	2.603	2.607	2.620	2.583	2.513	2.522	2.465	2.533	
	Ebn at Noon	311	304	293	277	259	250	260	276	289	304	305	314	
	Edh at Noon	35	32	29	24	21	19	20	24	30	33	36	35	
All-Sky Solar Radiation	RadAvg	2162	1856	1469	1005	677	534	602	878	1284	1707	2002	2245	
	RadStd	136	89	77	45	36	39	23	58	82	78	120	122	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	+0.56	N/A	N/A	+1.30	+0.74	N/A	N/A	-140	+167	+25

Nomenclature: See separate page